

Classification NAÏVE BAYES CLASSIFIER

Data Mining
Lecture # 14

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Bayes Classifier

- A probabilistic framework for solving classification problems
- Conditional Probability: $P(Y | X) = \frac{P(X, Y)}{P(X)}$
 $P(X | Y) = \frac{P(X, Y)}{P(Y)}$
- Bayes theorem: $P(Y | X) = \frac{P(X | Y)P(Y)}{P(X)}$

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Example of Bayes Theorem

- Given:
 - A doctor knows that meningitis causes stiff neck 50% of the time
 - Prior probability of any patient having meningitis is 1/50,000
 - Prior probability of any patient having stiff neck is 1/20
- If a patient has stiff neck, what's the probability he/she has meningitis?

$$P(M | S) = \frac{P(S | M)P(M)}{P(S)} = \frac{0.5 \times 1/50000}{1/20} = 0.0002$$

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Using Bayes Theorem for Classification

- Consider each attribute and class label as random variables
- Given a record with attributes $(X_1, X_2, \dots, X_d)$
 - Goal is to predict class Y
 - Specifically, we want to find the value of Y that maximizes $P(Y | X_1, X_2, \dots, X_d)$
- Can we estimate $P(Y | X_1, X_2, \dots, X_d)$ directly from data?

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Bayes Classifier

- A probabilistic framework for solving classification problems
 - calculate explicit probabilities for hypothesis
 - among the most practical approaches to certain types of learning problems
- Incremental:
 - each training example can incrementally increase/decrease the probability that a hypothesis is correct
 - prior knowledge can be combined with observed data
- A, C random variables
- Joint probability: $\Pr(A=a, C=c)$
- Conditional probability: $\Pr(C=c | A=a)$
- Relationship between joint and conditional probability distributions

$$\Pr(C, A) = \Pr(C | A) \times \Pr(A) = \Pr(A | C) \times \Pr(C)$$

$$\Pr(C | A) = \frac{P(A | C)P(C)}{P(A)}$$

Bayes Theorem:

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Example Data

Given a Test Record:

$X = (\text{Refund} = \text{No}, \text{Divorced}, \text{Income} = 120K)$

Id	Refund	Marital Status	Taxable income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	35K	Yes
6	No	Married	30K	No
7	Yes	Divorced	220K	No
8	No	Single	35K	Yes
9	No	Married	75K	No
10	No	Single	30K	Yes

- Can we estimate $P(\text{Evade} = \text{Yes} | X)$ and $P(\text{Evade} = \text{No} | X)$?

In the following we will replace
Evade = Yes by Yes, and
Evade = No by No

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Using Bayes Theorem for Classification

- Approach:
 - compute posterior probability $P(Y | X_1, X_2, \dots, X_d)$ using the Bayes theorem

$$P(Y | X_1, X_2, \dots, X_d) = \frac{P(X_1, X_2, \dots, X_d | Y) P(Y)}{P(X_1, X_2, \dots, X_d)}$$

- Maximum a-posteriori:** Choose Y that maximizes $P(Y | X_1, X_2, \dots, X_d)$
- Equivalent to choosing value of Y that maximizes $P(X_1, X_2, \dots, X_d | Y) P(Y)$
- How to estimate $P(X_1, X_2, \dots, X_d | Y)$?

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Example Data

Given a Test Record:

$X = (\text{Refund} = \text{No}, \text{Divorced}, \text{Income} = 120\text{K})$

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	30K	No
7	Yes	Divorced	220K	No
8	No	Single	35K	Yes
9	No	Married	75K	No
10	No	Single	30K	Yes

Using Bayes Theorem:

$$P(\text{Yes} | X) = \frac{P(X | \text{Yes}) P(\text{Yes})}{P(X)}$$

$$P(\text{No} | X) = \frac{P(X | \text{No}) P(\text{No})}{P(X)}$$

How to estimate $P(X | \text{Yes})$ and $P(X | \text{No})$?

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Naïve Bayes Classifier

- Assume independence among attributes X_i when class is given:
 - $P(X_1, X_2, \dots, X_d | Y_j) = P(X_1 | Y_j) P(X_2 | Y_j) \dots P(X_d | Y_j)$
- Now we can estimate $P(X_i | Y_j)$ for all X_i and Y_j combinations from the training data
- New point is classified to Y_j if $P(Y_j) \prod P(X_i | Y_j)$ is maximal.

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BAYESIAN LEARNING

Naïve (Simple) Bayesian Classification $P(C_i | X) = \frac{P(X | C_i) P(C_i)}{P(X)}$

Suppose that there are m classes $C_1, C_2, \dots, C_m$. Given an unknown data sample, X (i.e. having no class label), the classifier will predict that X belongs to the class having the highest posterior probability given X

Thus if $P(C_i | X) > P(C_j | X)$ for $1 \leq j \leq m, j \neq i$
then X is assigned to C_i

This is called Bayes decision rule i.e. Assign X to C_i if $P(X | C_i) P(C_i) > P(X | C_j) P(C_j)$ for $1 \leq j \leq m, j \neq i$
If $P(C_j)$ are unknown then assume that all $P(C_j)$ are equal to $1/m$

As $P(X)$ is constant for all classes, only $P(X | C_i) P(C_i)$ needs to be maximized.

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How to Estimate Probabilities from Data?

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	30K	No
7	Yes	Divorced	220K	No
8	No	Single	35K	Yes
9	No	Married	75K	No
10	No	Single	30K	Yes

• Class Prior Probability:

$$P(C = c) = \frac{N_c}{N}$$

e.g., $P(C = \text{No}) = 7/10$,
 $P(C = \text{Yes}) = 3/10$

• For discrete attributes:

$$P(A_i = a | C = c) = \frac{N_{a,c}}{N_c}$$

where $N_{a,c}$ is number of instances having attribute $A_i = a$ and belongs to class c

• Examples:

$$P(\text{Status} = \text{Married} | \text{No}) = 4/7$$

$$P(\text{Refund} = \text{Yes} | \text{Yes}) = 0$$

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Conditional Independence

- X and Y are conditionally independent given Z if $P(X|YZ) = P(X|Z)$
- Example: Arm length and reading skills
 - Young child has shorter arm length and limited reading skills, compared to adults
 - If age is fixed, no apparent relationship between arm length and reading skills
 - Arm length and reading skills are conditionally independent given age

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Naïve Bayes on Example Data

Given a Test Record:

$$X = (\text{Refund} = \text{No}, \text{Divorced}, \text{Income} = 120\text{K})$$

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

$$P(X | \text{Yes}) =$$

$$P(\text{Refund} = \text{No} | \text{Yes}) \times$$

$$P(\text{Divorced} | \text{Yes}) \times$$

$$P(\text{Income} = 120\text{K} | \text{Yes})$$

$$P(X | \text{No}) =$$

$$P(\text{Refund} = \text{No} | \text{No}) \times$$

$$P(\text{Divorced} | \text{No}) \times$$

$$P(\text{Income} = 120\text{K} | \text{No})$$

Estimate Probabilities from Data

$$\square \text{ Class: } P(Y) = N_c / N$$

$$\square \text{ e.g., } P(\text{No}) = 7/10, \\ P(\text{Yes}) = 3/10$$

$$\square \text{ For categorical attributes:}$$

$$P(X_i | Y_k) = |X_{ik}| / N_c$$

where $|X_{ik}|$ is number of instances having attribute value X_i and belonging to class Y_k

Examples:

$$P(\text{Status} = \text{Married} | \text{No}) = 4/7$$

$$P(\text{Refund} = \text{Yes} | \text{Yes}) = 0$$

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

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Estimate Probabilities from Data

□ For continuous attributes:

□ **Discretization:** Partition the range into bins:

- Replace continuous value with bin value
 - Attribute changed from continuous to ordinal

□ **Probability density estimation:**

- Assume attribute follows a normal distribution
- Use data to estimate parameters of distribution (e.g., mean and standard deviation)
- Once probability distribution is known, use it to estimate the conditional probability $P(X_i | Y)$

Estimate Probabilities from Data

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

□ Normal distribution:

$$P(X_i | Y_j) = \frac{1}{\sqrt{2\pi\sigma_{ij}^2}} e^{-\frac{(X_i - \mu_{ij})^2}{2\sigma_{ij}^2}}$$

□ One for each (X_i, Y_i) pair

□ For (Income, Class=No):

□ If Class=No

- sample mean = 110
- sample variance = 2975

$$P(\text{Income} = 120 | \text{No}) = \frac{1}{\sqrt{2\pi(54.54)}} e^{-\frac{(120-110)^2}{2(54.54)}} = 0.0072$$

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Example of Naïve Bayes Classifier

□ Creating a Naïve Bayes Classifier, essentially means to compute **counts**:

Total number of records: $N = 10$

Class No: Number of records: 7 Attribute Refund: Yes: 3 No: 4 Attribute Marital Status: Single: 2 Divorced: 1 Married: 4 Attribute Income: mean: 110 variance: 2975	Class Yes: Number of records: 3 Attribute Refund: Yes: 0 No: 3 Attribute Marital Status: Single: 2 Divorced: 1 Married: 0 Attribute Income: mean: 90 variance: 25
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Example of Naïve Bayes Classifier

Given a Test Record:

$$X = (\text{Refund} = \text{No}, \text{Divorced}, \text{Income} = 120\text{K})$$

Naïve Bayes Classifier:

$$\begin{aligned}
 P(\text{Refund} = \text{Yes} | \text{No}) &= 3/7 \\
 P(\text{Refund} = \text{No} | \text{No}) &= 4/7 \\
 P(\text{Refund} = \text{Yes} | \text{Yes}) &= 0 \\
 P(\text{Refund} = \text{No} | \text{Yes}) &= 1 \\
 P(\text{Marital Status} = \text{Single} | \text{No}) &= 2/7 \\
 P(\text{Marital Status} = \text{Divorced} | \text{No}) &= 1/7 \\
 P(\text{Marital Status} = \text{Married} | \text{No}) &= 4/7 \\
 P(\text{Marital Status} = \text{Single} | \text{Yes}) &= 2/3 \\
 P(\text{Marital Status} = \text{Divorced} | \text{Yes}) &= 1/3 \\
 P(\text{Marital Status} = \text{Married} | \text{Yes}) &= 0 \\
 P(X | \text{No}) &= P(\text{Refund} = \text{No} | \text{No}) \\
 &\quad \times P(\text{Divorced} | \text{No}) \\
 &\quad \times P(\text{Income} = 120\text{K} | \text{No}) \\
 &= 4/7 \times 1/7 \times 0.0072 = 0.0006 \\
 P(X | \text{Yes}) &= P(\text{Refund} = \text{No} | \text{Yes}) \\
 &\quad \times P(\text{Divorced} | \text{Yes}) \\
 &\quad \times P(\text{Income} = 120\text{K} | \text{Yes}) \\
 &= 1 \times 1/3 \times 1.2 \times 10^{-5} = 4 \times 10^{-6}
 \end{aligned}$$

For Taxable Income:
 If class = No: sample mean = 110
 sample variance = 2975
 If class = Yes: sample mean = 90
 sample variance = 25

Since $P(X|\text{No})P(\text{No}) > P(X|\text{Yes})P(\text{Yes})$
 Therefore $P(\text{No}|X) > P(\text{Yes}|X)$
 $\Rightarrow$ Class = No

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

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Example of Naïve Bayes Classifier

Given a Test Record:

$X = (\text{Refund} = \text{No}, \text{Divorced}, \text{Income} = 120\text{K})$

Naïve Bayes Classifier:

$P(\text{Refund} = \text{Yes} | \text{No}) = 3/7$
 $P(\text{Refund} = \text{No} | \text{No}) = 4/7$
 $P(\text{Refund} = \text{Yes} | \text{Yes}) = 0$
 $P(\text{Marital Status} = \text{Single} | \text{No}) = 2/7$
 $P(\text{Marital Status} = \text{Divorced} | \text{No}) = 1/7$
 $P(\text{Marital Status} = \text{Married} | \text{No}) = 4/7$
 $P(\text{Marital Status} = \text{Single} | \text{Yes}) = 2/3$
 $P(\text{Marital Status} = \text{Divorced} | \text{Yes}) = 1/3$
 $P(\text{Marital Status} = \text{Married} | \text{Yes}) = 0$

For Taxable Income:

If class = No: sample mean = 110
 sample variance = 2975
 If class = Yes: sample mean = 90
 sample variance = 25

Tid	Refund	Marital Status	Taxable Income	Class
1	Yes	Single	120K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	35K	Yes
6	No	Married	30K	No
7	Yes	Divorced	35K	Yes
8	No	Single	35K	Yes
9	No	Married	75K	No
10	No	Single	80K	Yes

$P(\text{Yes}) = 3/10$
 $P(\text{No}) = 7/10$
 $P(\text{Yes} | \text{Divorced}) = 1/3 \times 3/10 / P(\text{Divorced})$
 $P(\text{No} | \text{Divorced}) = 1/7 \times 7/10 / P(\text{Divorced})$
 $P(\text{Yes} | \text{Refund} = \text{No}, \text{Divorced}) = 1 \times 1/3 \times 3/10 / P(\text{Divorced}, \text{Refund} = \text{No})$
 $P(\text{No} | \text{Refund} = \text{No}, \text{Divorced}) = 4/7 \times 1/7 \times 7/10 / P(\text{Divorced}, \text{Refund} = \text{No})$

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Issues with Naïve Bayes Classifier

Naïve Bayes Classifier:

$P(\text{Refund} = \text{Yes} | \text{No}) = 3/7$
 $P(\text{Refund} = \text{No} | \text{No}) = 4/7$
 $P(\text{Refund} = \text{Yes} | \text{Yes}) = 0$
 $P(\text{Refund} = \text{No} | \text{Yes}) = 1$
 $P(\text{Marital Status} = \text{Single} | \text{No}) = 2/7$
 $P(\text{Marital Status} = \text{Divorced} | \text{No}) = 1/7$
 $P(\text{Marital Status} = \text{Married} | \text{No}) = 4/7$
 $P(\text{Marital Status} = \text{Single} | \text{Yes}) = 2/3$
 $P(\text{Marital Status} = \text{Divorced} | \text{Yes}) = 1/3$
 $P(\text{Marital Status} = \text{Married} | \text{Yes}) = 0$

For Taxable Income:
 If class = No: sample mean = 110
 sample variance = 2975
 If class = Yes: sample mean = 90
 sample variance = 25

$P(\text{Yes}) = 3/10$
 $P(\text{No}) = 7/10$

$P(\text{Yes} | \text{Married}) = 0 \times 3/10 / P(\text{Married})$
 $P(\text{No} | \text{Married}) = 4/7 \times 7/10 / P(\text{Married})$

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Issues with Naïve Bayes Classifier

Consider the table with Tid = 7 deleted

Tid	Refund	Marital Status	Taxable Income	Class
1	Yes	Single	120K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	35K	Yes
6	No	Married	30K	No
8	No	Single	35K	Yes
9	No	Married	75K	No
10	No	Single	80K	Yes

Given $X = (\text{Refund} = \text{Yes}, \text{Divorced}, 120\text{K})$

$P(X | \text{No}) = 2/6 \times 0 \times 0.0083 = 0$
 $P(X | \text{Yes}) = 0 \times 1/3 \times 1.2 \times 10^{-9} = 0$

Naïve Bayes Classifier:

$P(\text{Refund} = \text{Yes} | \text{No}) = 2/6$
 $P(\text{Refund} = \text{No} | \text{No}) = 4/6$
 $P(\text{Refund} = \text{Yes} | \text{Yes}) = 0$
 $P(\text{Refund} = \text{No} | \text{Yes}) = 1$
 $P(\text{Marital Status} = \text{Single} | \text{No}) = 2/6$
 $P(\text{Marital Status} = \text{Divorced} | \text{No}) = 0$
 $P(\text{Marital Status} = \text{Married} | \text{No}) = 4/6$
 $P(\text{Marital Status} = \text{Single} | \text{Yes}) = 2/3$
 $P(\text{Marital Status} = \text{Divorced} | \text{Yes}) = 1/3$
 $P(\text{Marital Status} = \text{Married} | \text{Yes}) = 0/3$
 For Taxable Income:
 If class = No: sample mean = 91
 sample variance = 685
 If class = Yes: sample mean = 90
 sample variance = 25

Naïve Bayes will not be able to classify X as Yes or No!

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Issues with Naïve Bayes Classifier

□ If one of the conditional probabilities is zero, then the entire expression becomes zero

□ Need to use other estimates of conditional probabilities than simple fractions

□ Probability estimation:

$$\text{Original: } P(A_i | C) = \frac{N_{ic}}{N_c}$$

$$\text{Laplace: } P(A_i | C) = \frac{N_{ic} + 1}{N_c + c}$$

$$\text{m - estimate: } P(A_i | C) = \frac{N_{ic} + mp}{N_c + m}$$

c : number of classes

p : prior probability of the class

m : parameter

N_c : number of instances in the class

N_{ic} : number of instances having attribute value A_i in class c

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Example of Laplacian correction

□ In order to avoid probability values of zero we can add one value to each category of the attribute with zero count category.

Examples 8.5:

Using the Laplacian correction to avoid computing probability values of zero. Suppose that for the class *buys.computer* = yes in some training database, D , containing 1000 tuples, we have 0 tuples with *income* = low, 990 tuples with *income* = medium, and 10 tuples with *income* = high. The probabilities of these events, without the Laplacian correction, are 0, 0.990 (from 990/1000), and 0.010 (from 10/1000), respectively. Using the Laplacian correction for the three quantities, we pretend that we have 1 more tuple for each income-value pair. In this way, we instead obtain the following probabilities (rounded up to three decimal places):

$$\frac{1}{1003} = 0.001, \quad \frac{991}{1003} = 0.988, \quad \text{and} \quad \frac{11}{1003} = 0.011,$$

respectively. The "corrected" probability estimates are close to their "uncorrected" counterparts, yet the zero probability value is avoided.

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Example of Naïve Bayes Classifier

Given a Test Record:

$X = (\text{Refund} = \text{No}, \text{Married}, \text{Income} = 120\text{K})$

With Laplace Smoothing

naïve Bayes Classifier:

$P(\text{Refund} = \text{Yes} | \text{No}) = 4/9$
 $P(\text{Refund} = \text{No} | \text{No}) = 5/9$
 $P(\text{Refund} = \text{Yes} | \text{Yes}) = 1/5$
 $P(\text{Refund} = \text{No} | \text{Yes}) = 4/5$
 $P(\text{Marital Status} = \text{Single} | \text{No}) = 3/10$
 $P(\text{Marital Status} = \text{Divorced} | \text{No}) = 2/10$
 $P(\text{Marital Status} = \text{Married} | \text{No}) = 5/10$
 $P(\text{Marital Status} = \text{Single} | \text{Yes}) = 3/6$
 $P(\text{Marital Status} = \text{Divorced} | \text{Yes}) = 2/6$
 $P(\text{Marital Status} = \text{Married} | \text{Yes}) = 1/6$

For taxable income:
 If class = No: sample mean = 110
 sample variance = 2975
 If class = Yes: sample mean = 90
 sample variance = 25

$P(X | \text{Class} = \text{No}) = P(\text{Refund} = \text{No} | \text{Class} = \text{No})$
 $\times P(\text{Married} | \text{Class} = \text{No})$
 $\times P(\text{Income} = 120\text{K} | \text{Class} = \text{No})$
 $= 5/9 \times 5/10 \times 0.0072$

$P(X | \text{Class} = \text{Yes}) = P(\text{Refund} = \text{No} | \text{Class} = \text{Yes})$
 $\times P(\text{Married} | \text{Class} = \text{Yes})$
 $\times P(\text{Income} = 120\text{K} | \text{Class} = \text{Yes})$
 $= 4/5 \times 1/6 \times 1.2 \times 10^{-4}$

$P(\text{No}) = 0.7, P(\text{Yes}) = 0.3$
 Since $P(X | \text{No}) P(\text{No}) > P(X | \text{Yes}) P(\text{Yes})$
 Therefore $P(\text{No} | X) > P(\text{Yes} | X)$
 $\Rightarrow \text{Class} = \text{No}$

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Naïve Bayesian classification – Example

□ Estimating $P(x|C)$

Outlook	
$P(\text{sunny} p) = 2/9$	$P(\text{sunny} n) = 3/5$
$P(\text{overcast} p) = 4/9$	$P(\text{overcast} n) = 0$
$P(\text{rain} p) = 3/9$	$P(\text{rain} n) = 2/5$

Temperature	
$P(\text{hot} p) = 2/9$	$P(\text{hot} n) = 2/5$
$P(\text{mild} p) = 4/9$	$P(\text{mild} n) = 2/5$
$P(\text{cool} p) = 3/9$	$P(\text{cool} n) = 1/5$

Humidity	
$P(\text{high} p) = 3/9$	$P(\text{high} n) = 4/5$
$P(\text{normal} p) = 6/9$	$P(\text{normal} n) = 1/5$

Windy	
$P(\text{true} p) = 3/9$	$P(\text{true} n) = 3/5$
$P(\text{false} p) = 6/9$	$P(\text{false} n) = 2/5$

 $P(p) = 9/14$ $P(n) = 5/14$

Outlook	Temperature	Humidity	Windy	Class
sunny	hot	high	false	N
sunny	hot	high	true	N
overcast	hot	high	false	P
rain	mild	high	false	P
rain	cool	normal	false	P
overcast	cool	normal	true	N
sunny	mild	high	false	N
sunny	cool	normal	false	P
rain	mild	normal	false	P
sunny	mild	normal	true	P
overcast	mild	high	true	P
hot	hot	normal	false	P
rain	mild	high	true	N

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Naïve Bayesian classification – Example

□ Classifying X:

□ an unseen sample $X = \langle \text{rain}, \text{hot}, \text{high}, \text{false} \rangle$ □ $P(X|p) \cdot P(p) =$

$$P(\text{rain}|p) \cdot P(\text{hot}|p) \cdot P(\text{high}|p) \cdot P(\text{false}|p) \cdot P(p) = 3/9 \cdot 2/9 \cdot 3/9 \cdot 6/9 \cdot 9/14 = 0.010582$$

□ $P(X|n) \cdot P(n) =$

$$P(\text{rain}|n) \cdot P(\text{hot}|n) \cdot P(\text{high}|n) \cdot P(\text{false}|n) \cdot P(n) = 2/5 \cdot 2/5 \cdot 4/5 \cdot 2/5 \cdot 5/14 = 0.018286$$

□ Sample X is classified in class n (don't play)

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Example of Naïve Bayes Classifier

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

A: attributes

M: mammals

N: non-mammals

$$P(A|M) = \frac{6}{7} \times \frac{6}{7} \times \frac{2}{7} \times \frac{2}{7} = 0.06$$

$$P(A|N) = \frac{1}{13} \times \frac{10}{13} \times \frac{3}{13} \times \frac{4}{13} = 0.0042$$

$$P(A|M)P(M) = 0.06 \times \frac{7}{20} = 0.021$$

$$P(A|N)P(N) = 0.004 \times \frac{13}{20} = 0.0027$$

 $P(A|M)P(M) > P(A|N)P(N)$
 $\Rightarrow$ Mammals

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

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X = <Red, SUV, Domestic> ?

Car	Color	Type	Origin	Stolen
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Imported	No
5	Yellow	Sports	Imported	Yes
6	Yellow	Sports	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Imported	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

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Car	Color	Type	Origin	Stolen
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Imported	No
5	Yellow	Sports	Imported	Yes
6	Yellow	Sports	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Imported	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

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Naïve Bayes (Summary)

- Robust to isolated noise points
- Handle missing values by ignoring the instance during probability estimate calculations
- Robust to irrelevant attributes
- Independence assumption may not hold for some attributes
 - Use other techniques such as Bayesian Belief Networks (BBN)
- Naïve Bayes can produce a probability estimate, but it is usually a very biased one
 - Logistic Regression is better for obtaining probabilities.

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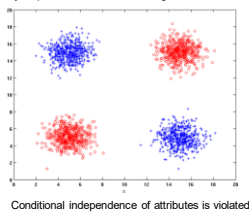
Naïve Bayes (Summary)

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Naïve Bayes

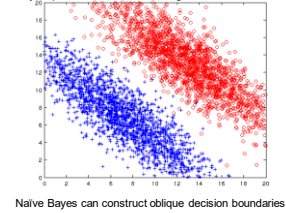
- How does Naïve Bayes perform on the following dataset?



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Naïve Bayes

- How does Naïve Bayes perform on the following dataset?



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Naïve Bayes

- How does Naïve Bayes perform on the following dataset?

Y = 1	1	1	1	0
Y = 2	0	1	0	0
Y = 3	0	0	1	1
Y = 4	0	0	1	1
	X = 1	X = 2	X = 3	X = 4

Conditional independence of attributes is violated

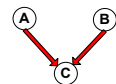
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Bayesian Belief Networks

- Provides graphical representation of probabilistic relationships among a set of random variables

- Consists of:

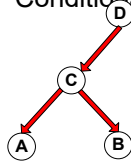
- A directed acyclic graph (dag)
 - Node corresponds to a variable
 - Arc corresponds to dependence relationship between a pair of variables



- A probability table associating each node to its immediate parent

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Conditional Independence



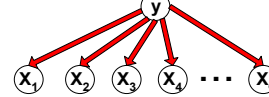
D is parent of C
A is child of C
B is descendant of D
D is ancestor of A

- A node in a Bayesian network is conditionally independent of all of its nondescendants, if its parents are known

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Conditional Independence

- Naïve Bayes assumption:



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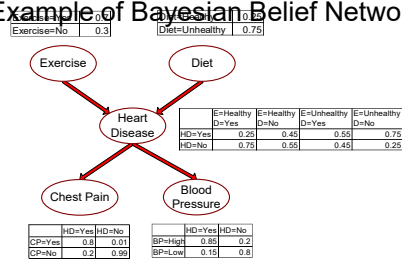
Probability Tables

- If X does not have any parents, table contains prior probability $P(X)$
- If X has only one parent (Y), table contains conditional probability $P(X|Y)$
- If X has multiple parents ($Y_1, Y_2, \dots, Y_k$), table contains conditional probability $P(X|Y_1, Y_2, \dots, Y_k)$



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Example of Bayesian Belief Network



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Example of Inferencing using BBN

- Given: $X = (E=\text{No}, D=\text{Yes}, CP=\text{Yes}, BP=\text{High})$
- Compute $P(HD|E, D, CP, BP)?$
- $P(HD=\text{Yes} | E=\text{No}, D=\text{Yes}) = 0.55$
 $P(CP=\text{Yes} | HD=\text{Yes}) = 0.8$
 $P(BP=\text{High} | HD=\text{Yes}) = 0.85$
- $P(HD=\text{Yes} | E=\text{No}, D=\text{Yes}, CP=\text{Yes}, BP=\text{High})$
 $\propto 0.55 \times 0.8 \times 0.85 = 0.374$
- $P(HD=\text{No} | E=\text{No}, D=\text{Yes}) = 0.45$
 $P(CP=\text{Yes} | HD=\text{No}) = 0.01$
 $P(BP=\text{High} | HD=\text{No}) = 0.2$
- $P(HD=\text{No} | E=\text{No}, D=\text{Yes}, CP=\text{Yes}, BP=\text{High})$
 $\propto 0.45 \times 0.01 \times 0.2 = 0.0009$

Classify X
as Yes

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